INCH-POUND

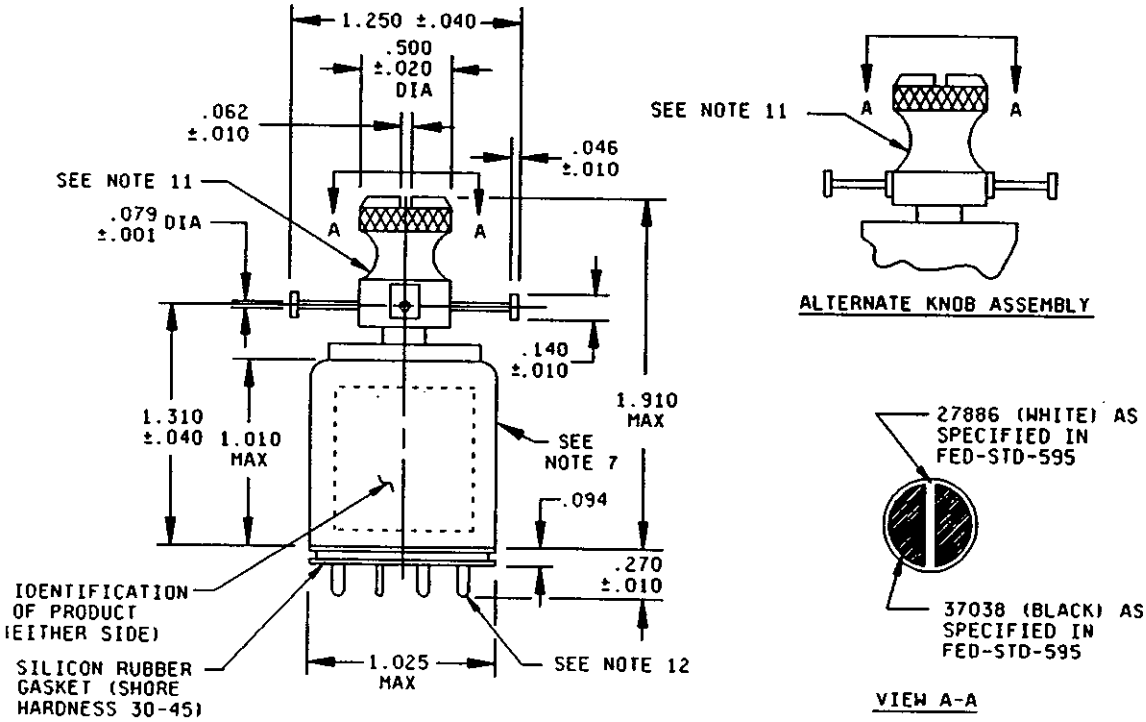
MIL-R-6106/37D(USAF)
17 February 1993
SUPERSEDING
MIL-R-6106/37C(USAF)
28 March 1990

MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, TYPE I, 4 PDT, 10 AMPERES, MAGNETIC LATCH, PERMANENT MAGNET DRIVE, HERMETICALLY SEALED ASSEMBLY

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-6106.

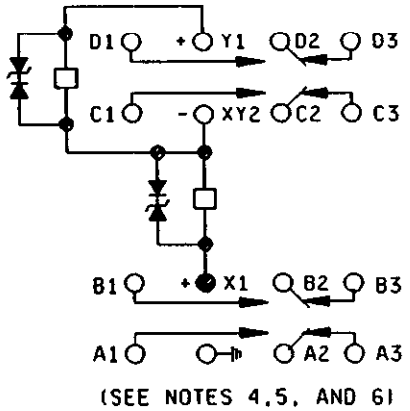


(D) FIGURE 1. Relay dimensions and configuration.

(D) denotes changes

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CIRCUIT DIAGRAM

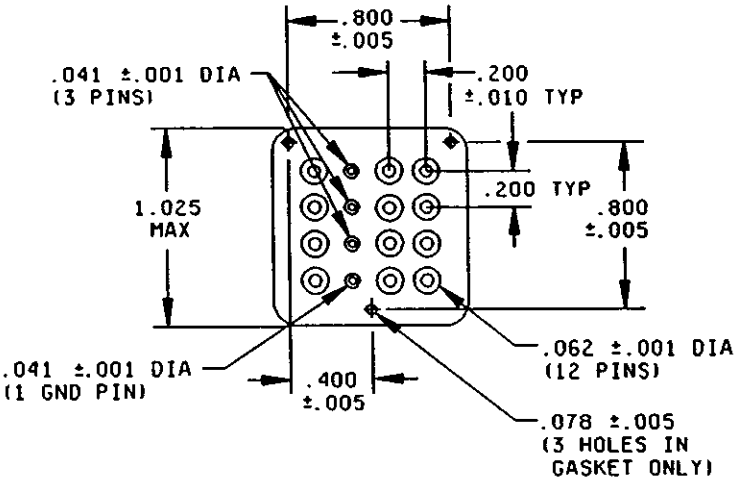


FIGURE 1. Relay dimensions and configuration - Continued.

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Inches	mm	Inches	mm
.001	0.03	.200	5.08
.005	0.13	.270	6.86
.010	0.25	.400	10.16
.020	0.51	.500	12.70
.040	1.02	.800	20.32
.041	1.04	1.010	25.65
.062	1.57	1.025	26.04
.078	1.98	1.250	31.75
.079	2.01	1.310	33.27
.094	2.39	1.910	48.51
.140	3.56		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 inch (0.25 mm) for three place decimals and ±.03 inch (0.8 mm) for two place decimals.
4. Relay is magnetically latched in both positions.
5. Caution note to observe polarity must appear on relays with dc coils.
6. Terminal numbers shall not appear on the header or on the gasket. There shall be affixed to the relay a legible circuit diagram that identifies each terminal location specified.
7. Enclosure shall be brown 30252 in accordance with FED-STD-595.
8. With the same magnitude and polarity voltage applied to terminals X1-XY2 and Y1-XY2 simultaneously, the relay contacts shall not transfer from the condition existing prior to the application of voltage to both coils.
9. With 20 V dc maximum applied initially to terminals X1-XY2 the subsequent application of 28 V dc minimum to terminals Y1-XY2 (while the voltage is maintained on terminals X1-XY2) shall not cause the relay contacts to transfer from the condition resulting from the application of voltage to terminals X1-XY2 prior to the application of voltage to terminals Y1-XY2. Removal of voltage from terminals X1-XY2 (while voltage is maintained on terminals Y1-XY2) shall cause the relay contacts to transfer.
10. With 20 V dc maximum applied initially to terminals Y1-XY2 the subsequent application of 28 V dc to terminals X1-XY2 (while the voltage is maintained on terminals Y1-XY2) shall not cause the relay contacts to transfer from the condition resulting from the application of voltage to terminals Y1-XY2 prior to the application of voltage to terminals X1-XY2. Removal of voltage from terminals Y1-XY2 (while the voltage is maintained on terminals X1-XY2) shall cause the relay contacts to transfer.
11. Knob and post shall be nickel plated (100 microinches minimum) 303 stainless steel.
12. All terminals shall be finished with gold plate in accordance with MIL-G-45204, type II, class 1. Underplating shall be nickel, 50 to 150 microinches thick.

FIGURE 1. Relay dimensions and configuration - Continued.

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REQUIREMENTS:

Coil data: See table I.

Rated contact load: See table II.

Weight: 0.2 pound maximum.

Temperature range: -70°C to +125°C.

Maximum altitude rating: 80,000 feet.

Shock: MIL-STD-202, method 213, test condition A 1/:

G-level: 50 g's.

Duration: 10 ms.

Maximum duration contact opening: 10 microseconds.

Maximum duration contact closing: 1 microsecond.

Vibration-sinusoidal 1/:

G-level: 30 g's.

Frequency range: 10 to 3,000 Hz.

Vibration scan: Applicable to group A testing.

Frequency range: 5 to 2,000 Hz.

Intensity: 0.01 double inch amplitude or ±2 g's, whichever is less.

Vibration-random 1/: MIL-STD-202, method 214, test condition IG, (0.4 g^2/Hz), 15 minutes each plane, 50 to 2,000 Hz.

Shock: Test condition C.

Acceleration 2/: 15 g's.

Insulation resistance, initial: 100 megohms.

After life or environmental tests: 100 megohms.

Dielectric withstanding voltage (sea level):

	Initial	After life tests
Coil to case	1050	1050
Aux contacts	---	---
All other points	1500	1250

Dielectric withstanding voltage (altitude):

	Initial	After life tests
Coil to case	350	350
Aux contacts	---	---
All other points	350	350

1/ To be tested as complete relay assembly, see table I, footnote 1/.
2/ To be tested with relay inserted in socket, MIL-R-6106/5-006.

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Maximum contact drop:

Initial: 0.150 volt.

After life test: 0.175 volt.

Duty rating: Continuous.

Back EMF (coil transient): ±42 volts maximum.

Test temperature: Room ambient.

Point of measurement: Across coil terminals X1-XY2 and Y1-XY2.

Coil voltage: 28 V dc.

Test method: Apply coil voltage to X1-XY2. With oscilloscope monitoring the coil terminals, measure and record the back electromotive force when voltage is removed. Same procedure applies for Y1-XY2.

Operational reliability: Applicable.

Identification of product: Applicable. The circuit diagram shall be oriented so that when the relay is held with the circuit diagram right side up as shown on figure 1 and rotated away from the viewer about a horizontal axis through the diagram until the header terminals face the viewer, then each terminal shall be in the location shown in the circuit diagram.

Part or Identifying Number (PIN): M6106/37-001.

(D) TABLE I. Dash number and operating characteristics. 1/

	Coil data							Time-milliseconds - max							
	Rated		Max		Max pick-up voltage					Bounce					
										Main		Aux			
Dash no.	V dc 2/ 3/	Res Ω min	V dc	Am-pere	Nom-inal (V dc) 4/	High temp test (V dc)	Cont cur-rent test (V dc)	Oper-ate 5/	Reset 5/	NO	NC	NO	NC	Over-load cur-rent (am-peres)	Rup-ture cur-rent (am-peres)
001	28 3/	350	30	0.08	18.0	20	22.5	15	15	1	1	---	---	40	50

1/ The complete relay assembly shall consist of the following:
a. Required relay.
b. Bracket and socket assembly (see MIL-R-6106/5).
c. Mounting track (see MIL-R-6106/3).

(D) 2/ CAUTION: The use of any coil voltage less than rated coil voltage will compromise the operation of the relay.

3/ Equipped with back EMF (coil transient) suppression. JANTX or equivalent screened semiconductors shall be used. Relays using suppression devices shall continue to operate should the suppression circuit be in a failure mode.

4/ Over the temperature range.

(D) 5/ With rated coil voltage.

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TABLE II. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase			
		Main		Aux		Main		Aux		Main		Aux	
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz
Resistive	100	10	10	---	---	10	---	---	---	10	---	---	---
Inductive	20	8	8	---	---	8	---	---	---	8	---	---	---
Inductive	100	(Not applicable)											
Motor	100	4	4	---	---	4	---	---	---	4	---	---	---
Lamp	100	2	2	---	---	2	---	---	---	2	---	---	---
Transfer, load													
Mechanical life (reduced load)	400	2.5	2.5	---	---	2.5	---	---	---	2.5	---	---	---
Inter-mediate current	(Applicable per MIL-R-6106)												

CONCLUDING MATERIAL

Custodian:
Air Force - 85

Review activity:
Air Force - 99

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-F735-01)

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